University of Moratuwa

Department of Electronic and Telecommunication Engineering

EN2160: Electronic Design Realization

Report- Preliminary Design Part

Index No: 200396U

Name: Miranda C.M.C.C.

Product: Smart Plug

Implemented Design

Sketches

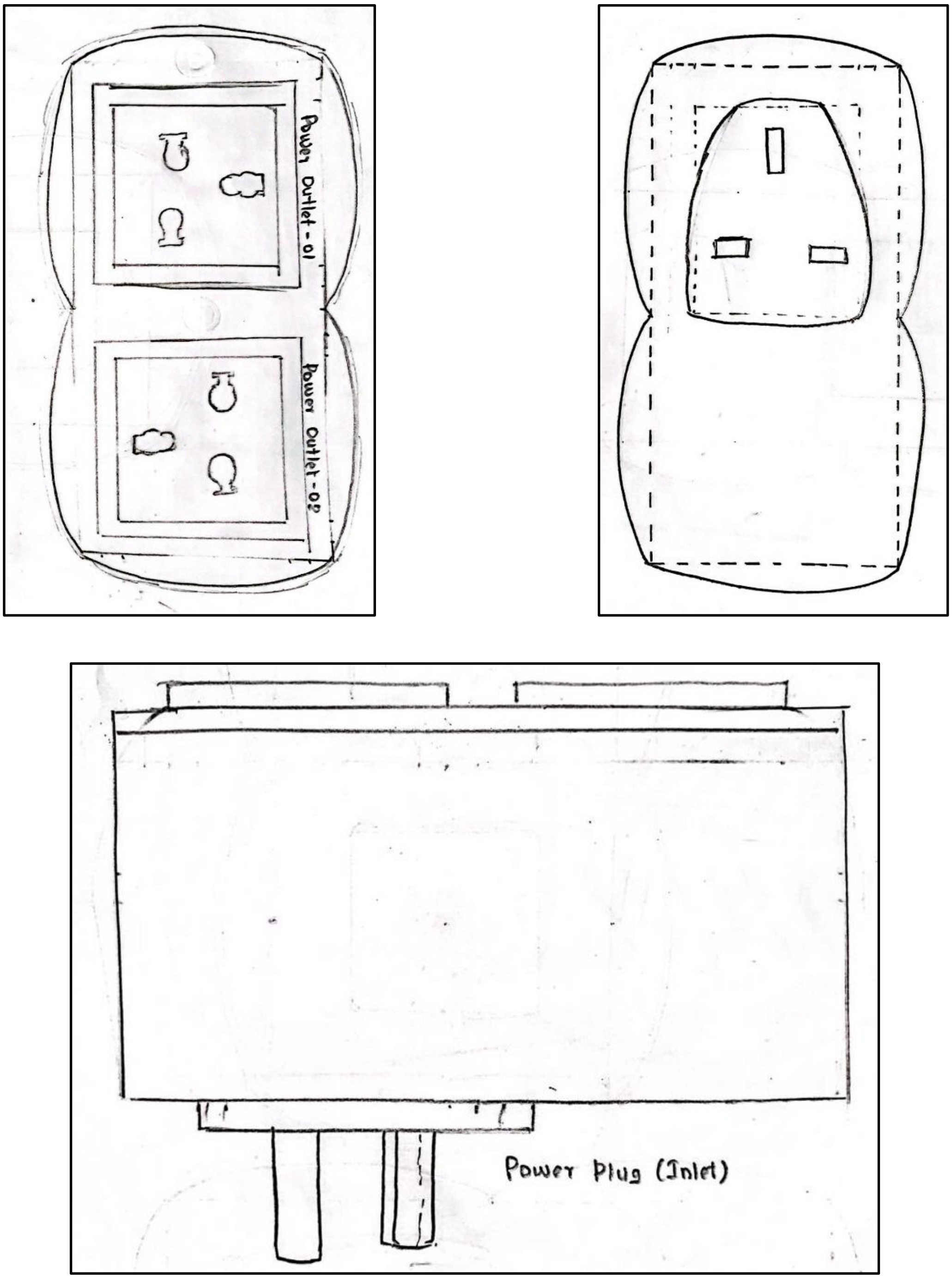


Figure 1: Sketches: Initial Design

Solidworks Designs

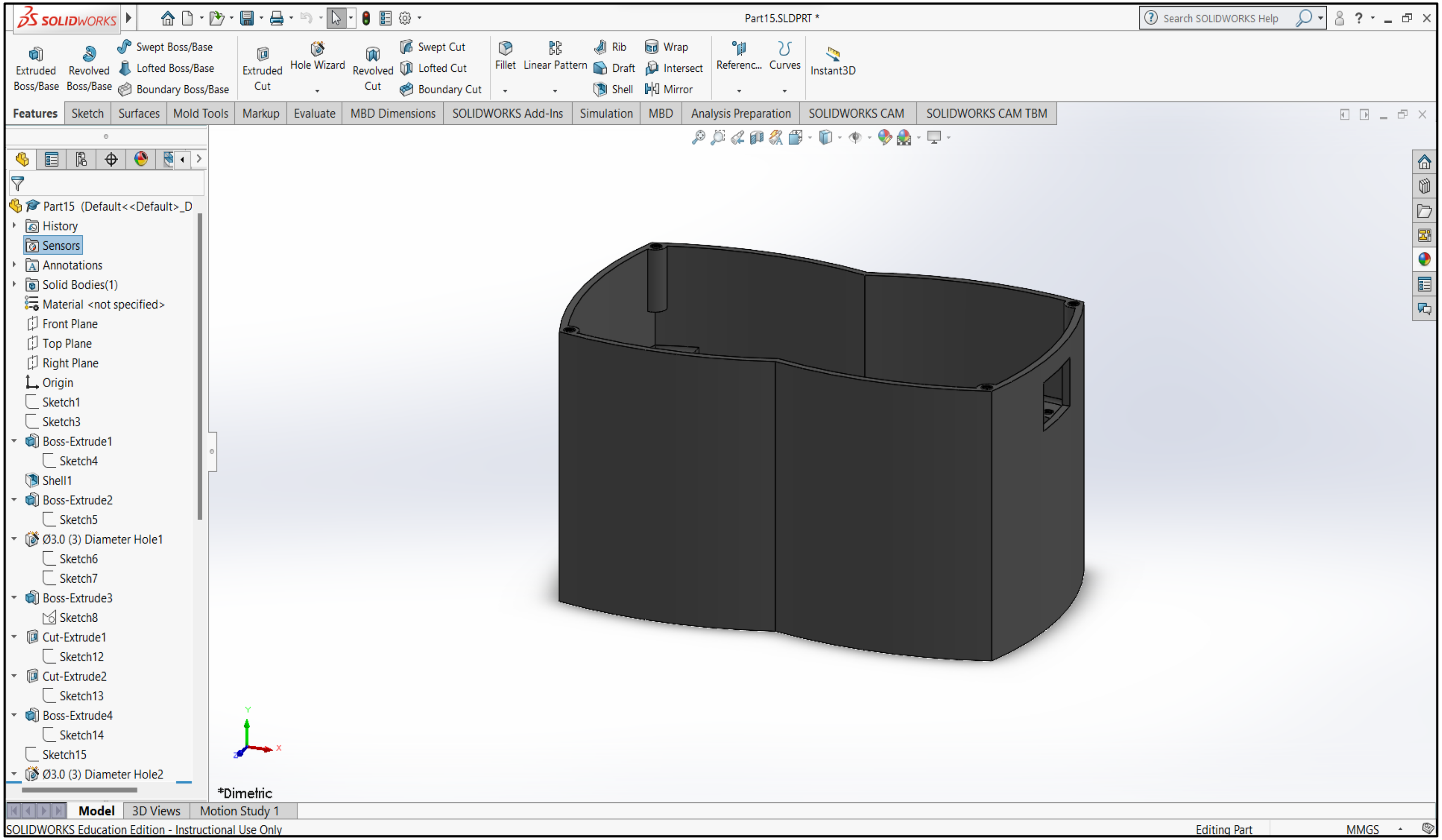


Figure 2: Solidworks Design: Initial (1 of 3)

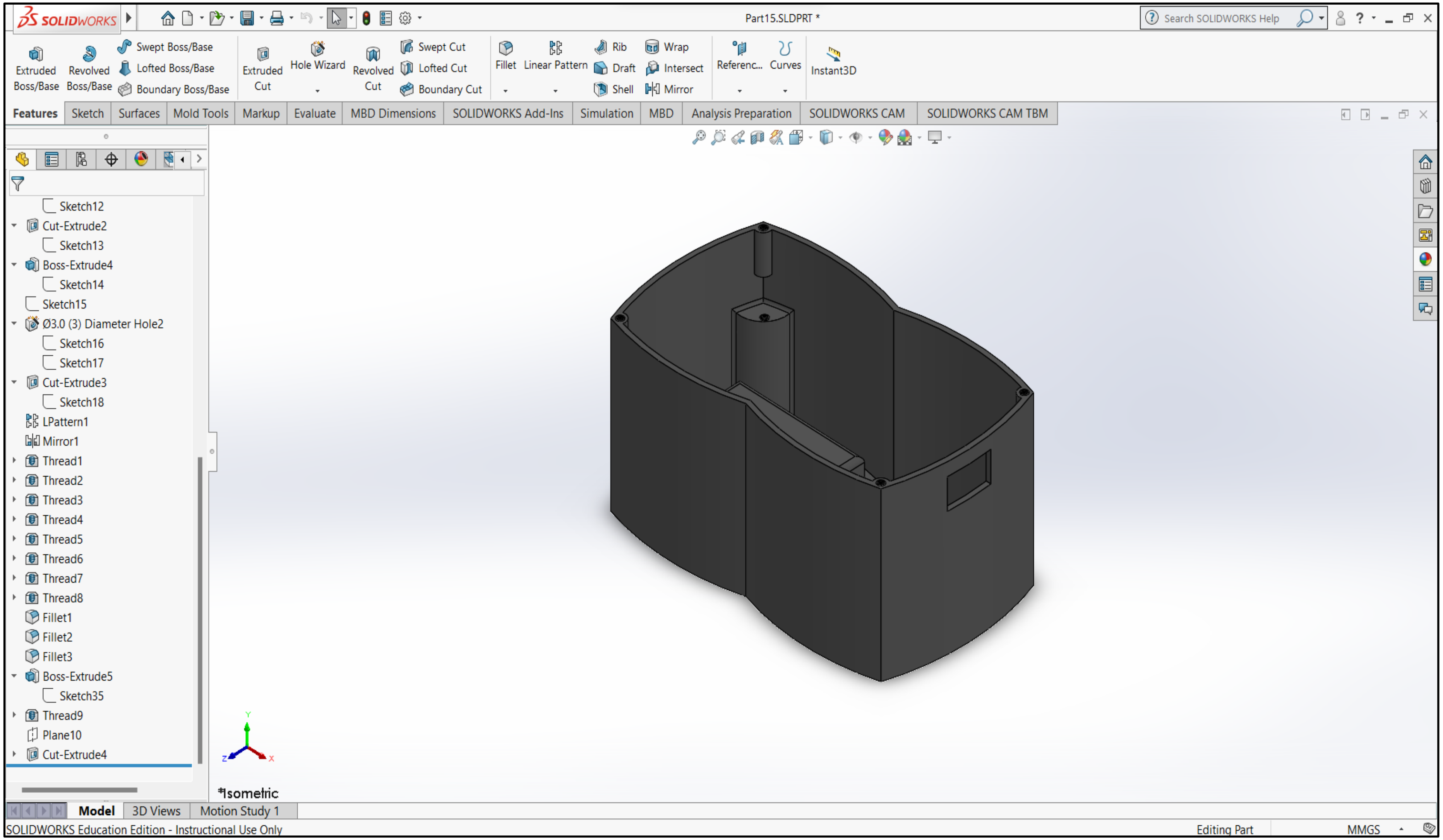


Figure 3: Solidwork Design: Initial (2 of 3)

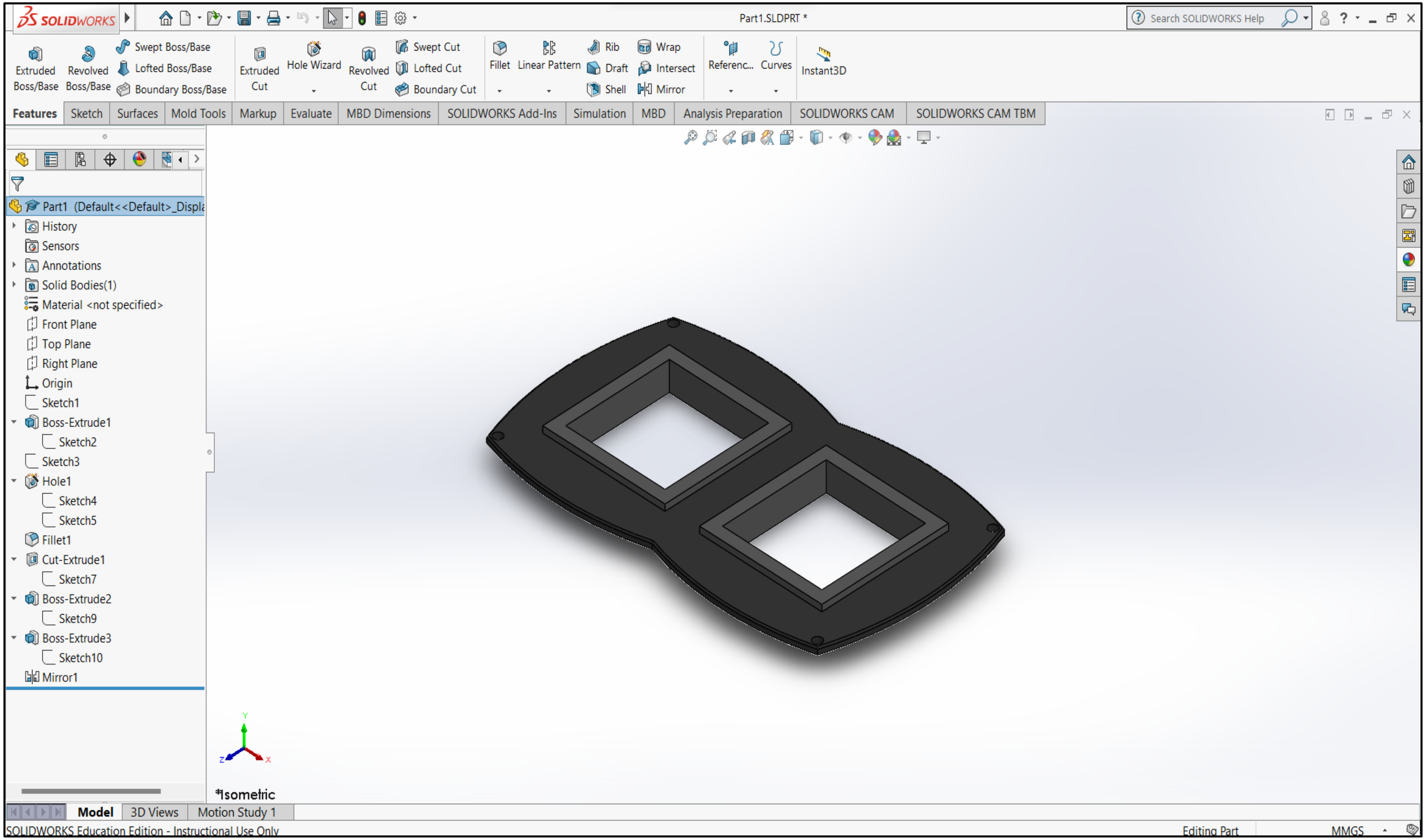


Figure 4: Solidworks Design: Initial (3 of 3)

Block Diagram

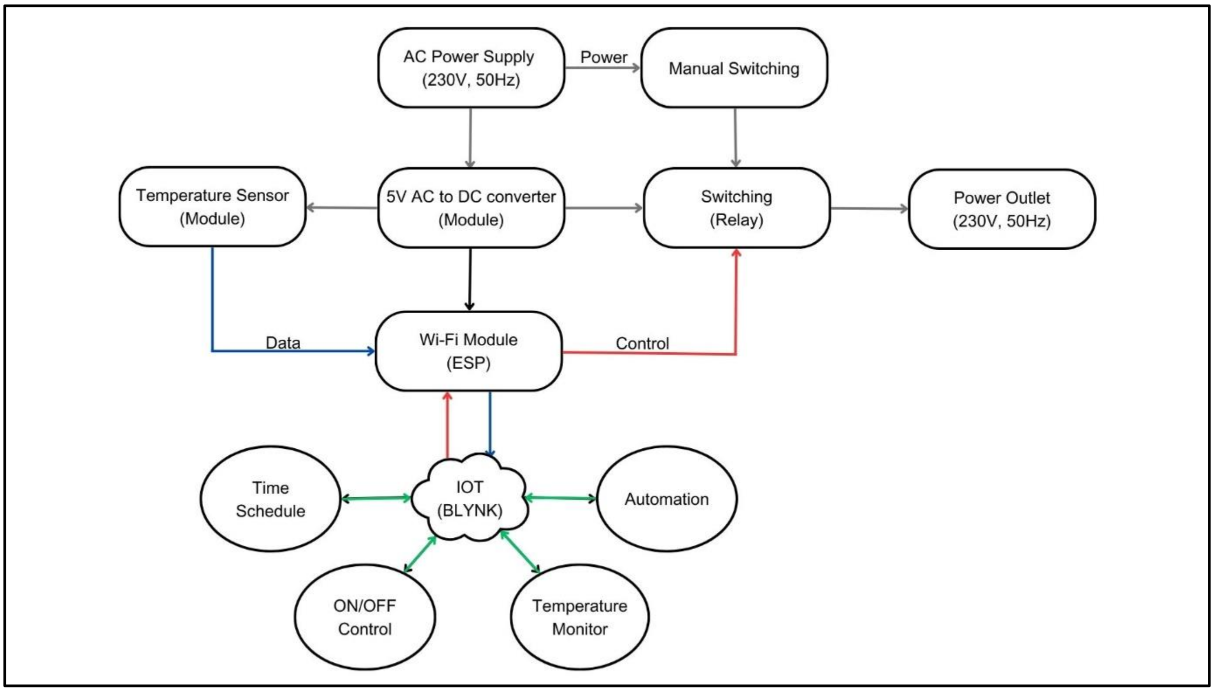


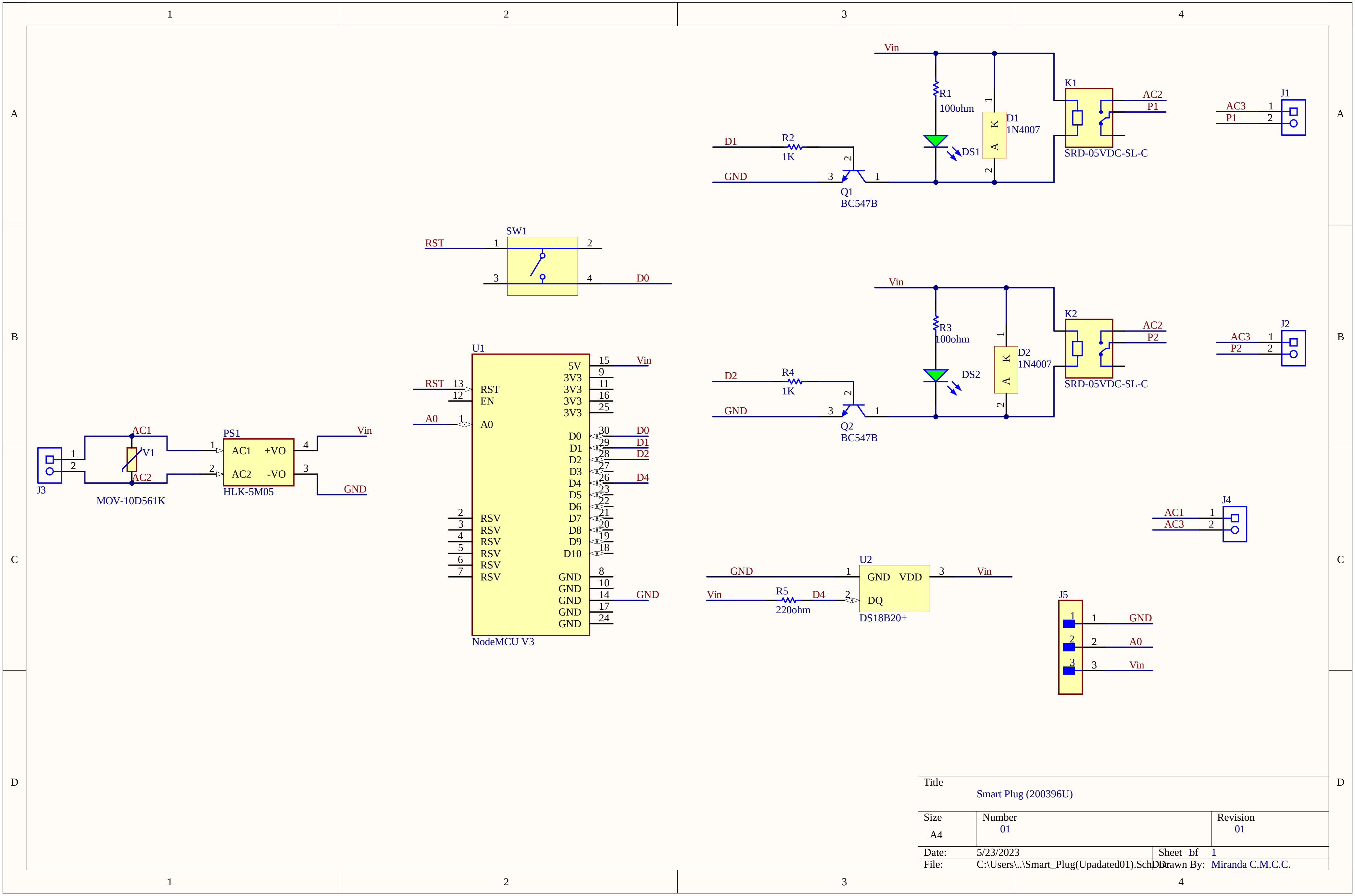
Figure 5: Block Diagram (Initial Design)

The smart plug was initially developed with a conventional design, consisting of two power outlets and the capability to be wirelessly controlled through a Wi-Fi connection.

The mentioned block diagram outlines the various features and specifications of this smart plug.

These are the features that the IoT platform will provide,

- There will be two distinct switch controls provided to manage the individual outlets.
- The platform allows for the scheduling of ON/OFF times for the smart plug.
- Utilizing the temperature data obtained from the sensor, the smart plug will incorporate an automation feature that automatically switches off when the temperature surpasses the upper threshold level. Likewise, it turns on the plug automatically once the temperature drops below the lower threshold level.



Vin
R1
100ohm
D1
1N4007
K1
SRD-05VDC-SL-C
AC2
P1
J1
AC3
D1
R2
1K
DS1
GND
Q1
BC547B
SW1
RST
D0
Vin
R3
100ohm
D2
1N4007
K2
SRD-05VDC-SL-C
AC2
P2
J2
AC3
P2
D2
R4
1K
DS2
GND
Q2
BC547B
U1
NodeMCU V3
RST
EN
A0
5V
3V3
D0
D1
D2
D3
D4
D5
D6
D7
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D4
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A0
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Size
A4
Number
01
Revision
01
Date: 5/23/2023
Sheet 1 of 1
File: C:\Users\..\Smart_Plug(Upadated01).SchDoc
Drawn By: Miranda C.M.C.C.

Problems identified after considering the course content.

- **Mass production:**
    - When it comes to producing a large quantity of units, 3D printing is not the most efficient approach. In contrast, molding enables quicker and more cost-effective production of identical components. However, my previous design cannot undergo the molding process, indicating that it might have features or geometries that are incompatible with molding.

- **Draft angles:**
    - In the process of molding, draft angles hold significant importance as they facilitate the effortless extraction of the molded component from the mold. Insufficient draft angles can lead to part entrapment or subpar surface finishing. To ensure seamless production and the creation of high-quality molded parts, it is crucial to integrate appropriate draft angles into the design. Unfortunately, I neglected to consider the importance of draft angles in my previous design.

- **Visual aesthetics:**
    - The visual appeal of a product plays a vital role in captivating customers and creating a desirable product. While functionality remains significant, it is equally important to consider the overall visual appeal of the design. Elements such as sleek lines, well-balanced proportions, and an aesthetically pleasing color palette should be considered to enhance the attractiveness of the product. Regrettably, I failed to consider this aspect in my previous design.

- **Screw positioning:**
    - For improved market appeal and user satisfaction, it is advisable to conceal screws from direct view. Placing screws on the underside or within the enclosure contributes to a neater and more polished aesthetic. However, in my enclosure design, I positioned the screws on the top surface, which detracted from its overall cleanliness and professional appearance.

- **Surface modeling:**
    - Surface modeling encompasses the creation of sleek and visually pleasing curves, chamfers, and fillets on the outer surfaces of the enclosure. This method is frequently employed to achieve a sophisticated and premium aesthetic. Unfortunately, I did not incorporate this approach in my previous design, resulting in a lack of professional and refined appearance.

- **Enclosure assembly:**
    - To eliminate the reliance on adhesives such as hot glue, it is crucial to achieve seamless integration of the enclosure components, eliminating the necessity for supplementary fixes. This necessitates careful scrutiny of details and meticulous dimensional planning. However, I made certain errors in this aspect as well.

- **Hand-drawn Sketches**
    - Neglecting the recommended procedure, I deviated from the ideal approach for enclosure design by not commencing with hand-drawn sketches.

I have implemented the essential enhancements in my design to fulfill the requirements. I have commenced the enclosure design process with hand-drawn sketches. The design now possesses compatibility for molding, exhibits aesthetic appeal, integrates suitable draft angles, conceals screws, employs surface modeling techniques, and guarantees secure enclosure assembly without the necessity of adhesives.

Problems/Improvements identified/proposed by members of group.

Problems

- **Insufficient presence of visually pleasing curves:**
    - The initial design lacked the inclusion of attractive curves. To rectify this, my group members proposed integrating a greater number of curves into the enclosure design, thereby enhancing its overall visual appeal.

- **Limitations in wiring space:**
    - The previous design faced constraints in terms of available space, posing challenges in accommodating appropriate wiring within the product. To overcome this issue, my teammates recommended making design modifications that would allow for sufficient space to facilitate proper wiring connections.

Improvements

- **Separate power switches**
    - In the initial design, a single power switch was implemented to manually control both power outlets simultaneously. However, based on the recommendations of my group members, it was advised to adopt a revised approach by incorporating two separate power switches, each dedicated to controlling an individual power outlet.

Problems/Improvements identified/proposed by users.

Based on user feedback obtained through a comprehensive user survey, I identified various issues in my design and schematic. Consequently, I made necessary modifications and implemented improvements as suggested by the users.

Problems

- **No USB outlets for direct charging**
  - In the initial design, the absence of USB outlets for direct device charging without a power adapter was a notable concern raised by several users. As a result, there was a collective request for the inclusion of both USB-C and USB-A outlets to address this issue.

Improvements

- **Power monitoring feature**
  - Based on user feedback, there was a strong desire for a power monitoring feature within the mobile application. This functionality would enable users to track their power consumption, empowering them to effectively monitor and plan their activities accordingly.

- **Voice Control feature**
  - A subset of users indicated a preference for the inclusion of voice control functionality alongside the existing remote-control capabilities through the mobile application. This enhancement would significantly enhance user convenience and simplify their daily lives.

I have diligently implemented all the recommended improvements provided by both my group members and the users. I have exerted considerable effort to address the design issues. To incorporate USB outlets, I utilized a power bank module, while integrating voice control capabilities through various platforms. Additionally, I incorporated a current sensor module to enable power monitoring, enhancing the IoT platform to display this information to users. Furthermore, I have made significant enhancements to the enclosure design, ensuring adequate spacing for wiring requirements.

Improved Design

Sketches

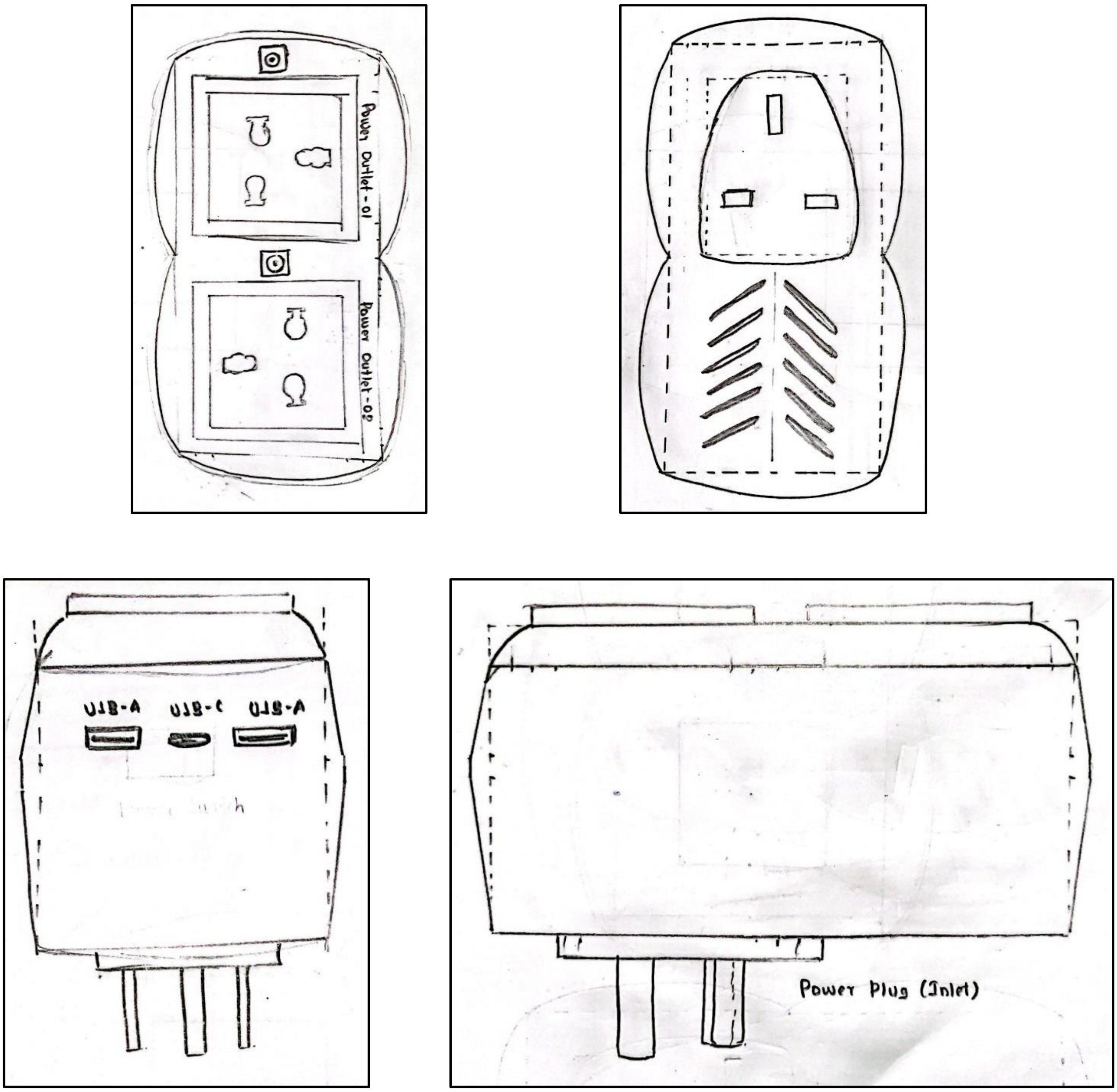


Figure 6: Sketches (Improved Design)

Solidworks Designs

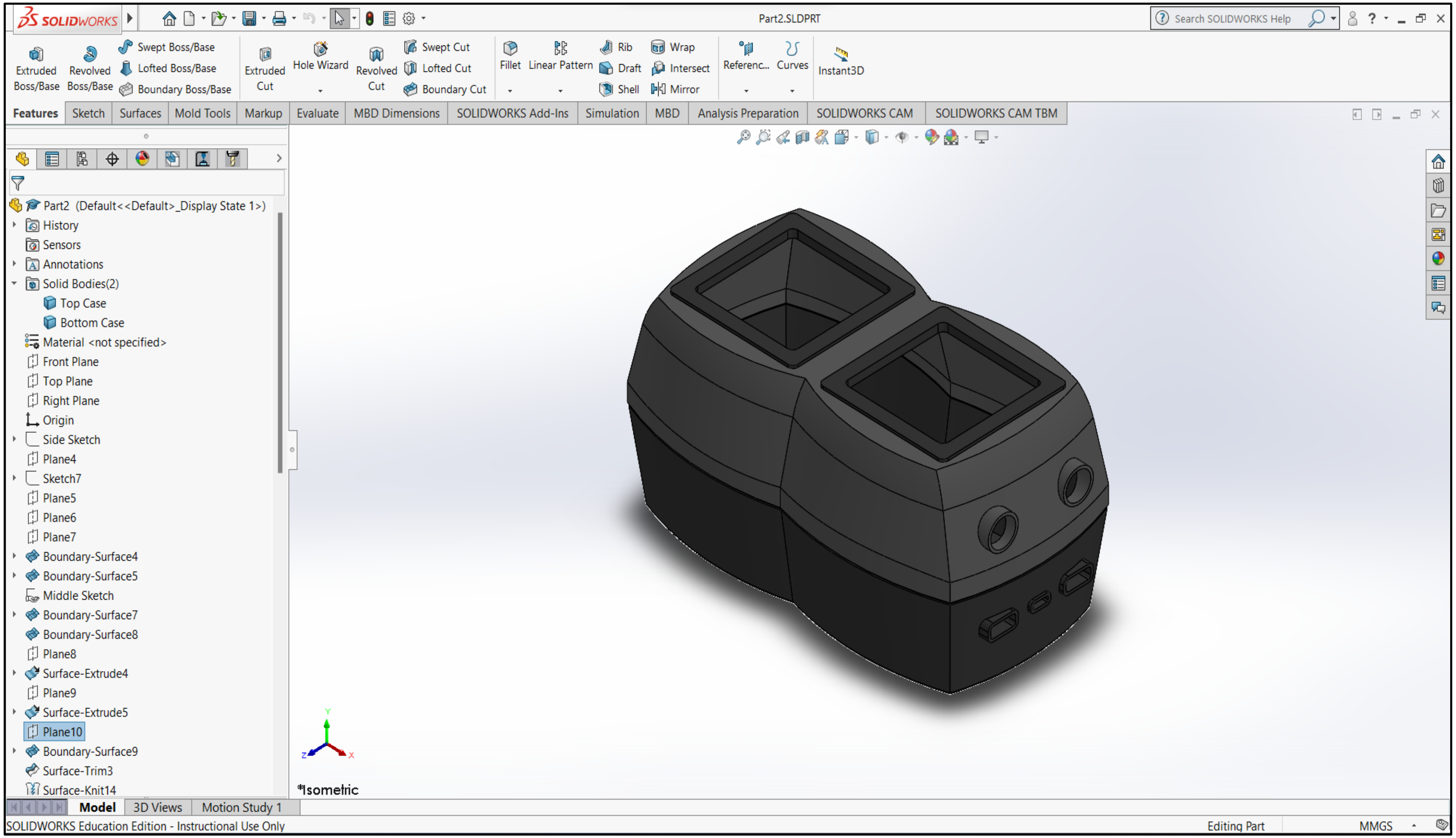


Figure 7: Solidworks Design: Improved (1 of 2)

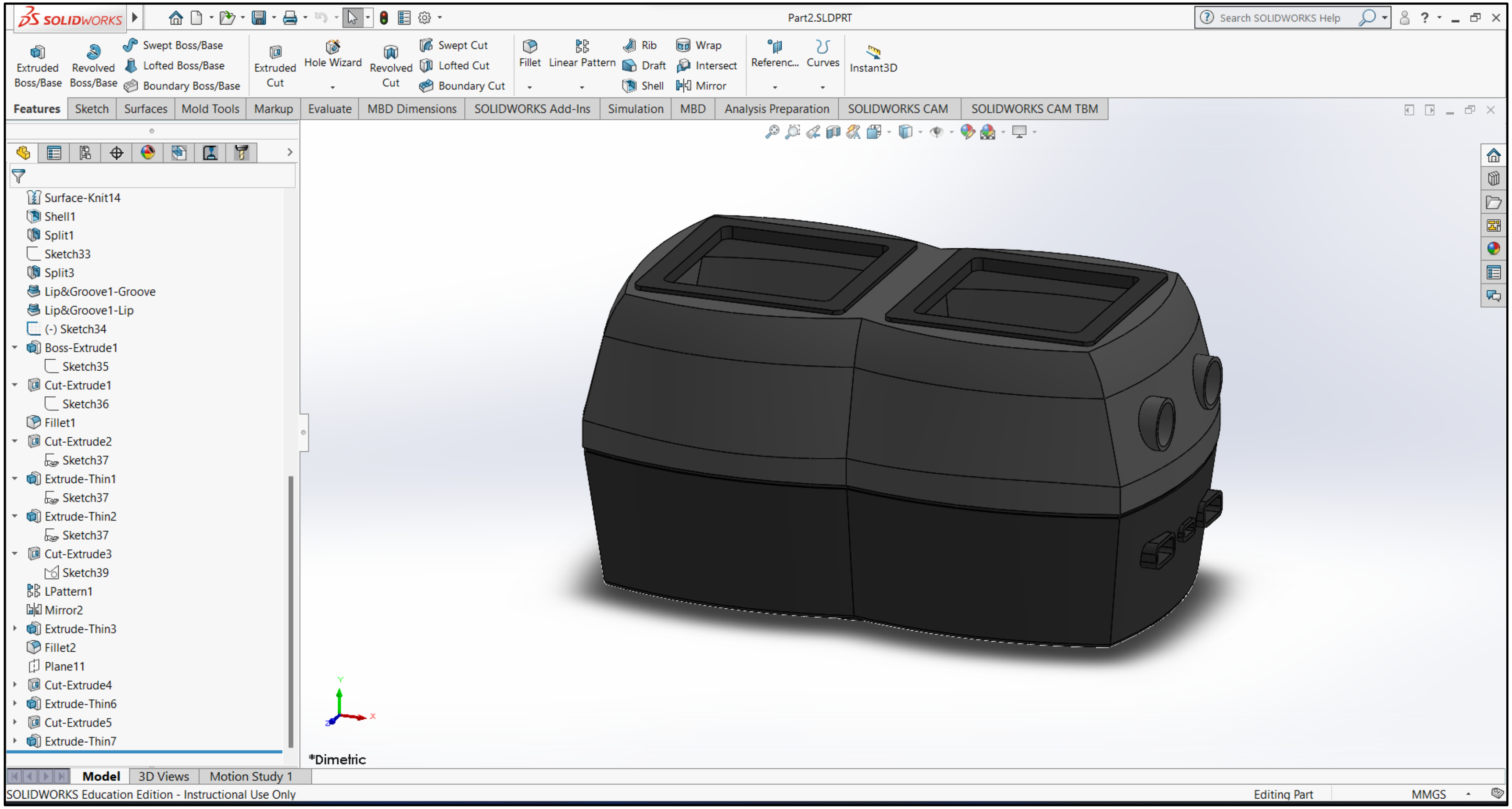


Figure 8: Solidworks Design: Improved (2 of 2)

Block Diagram

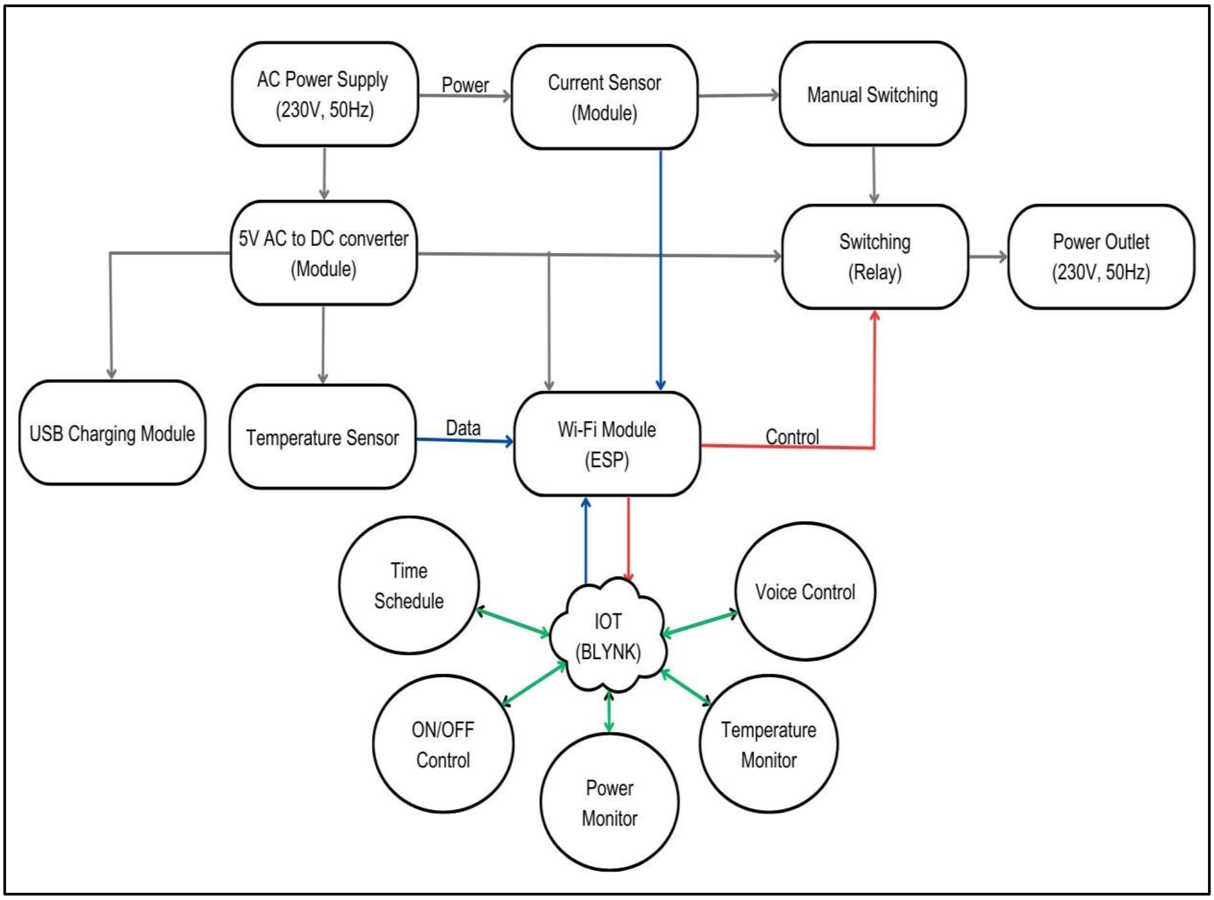
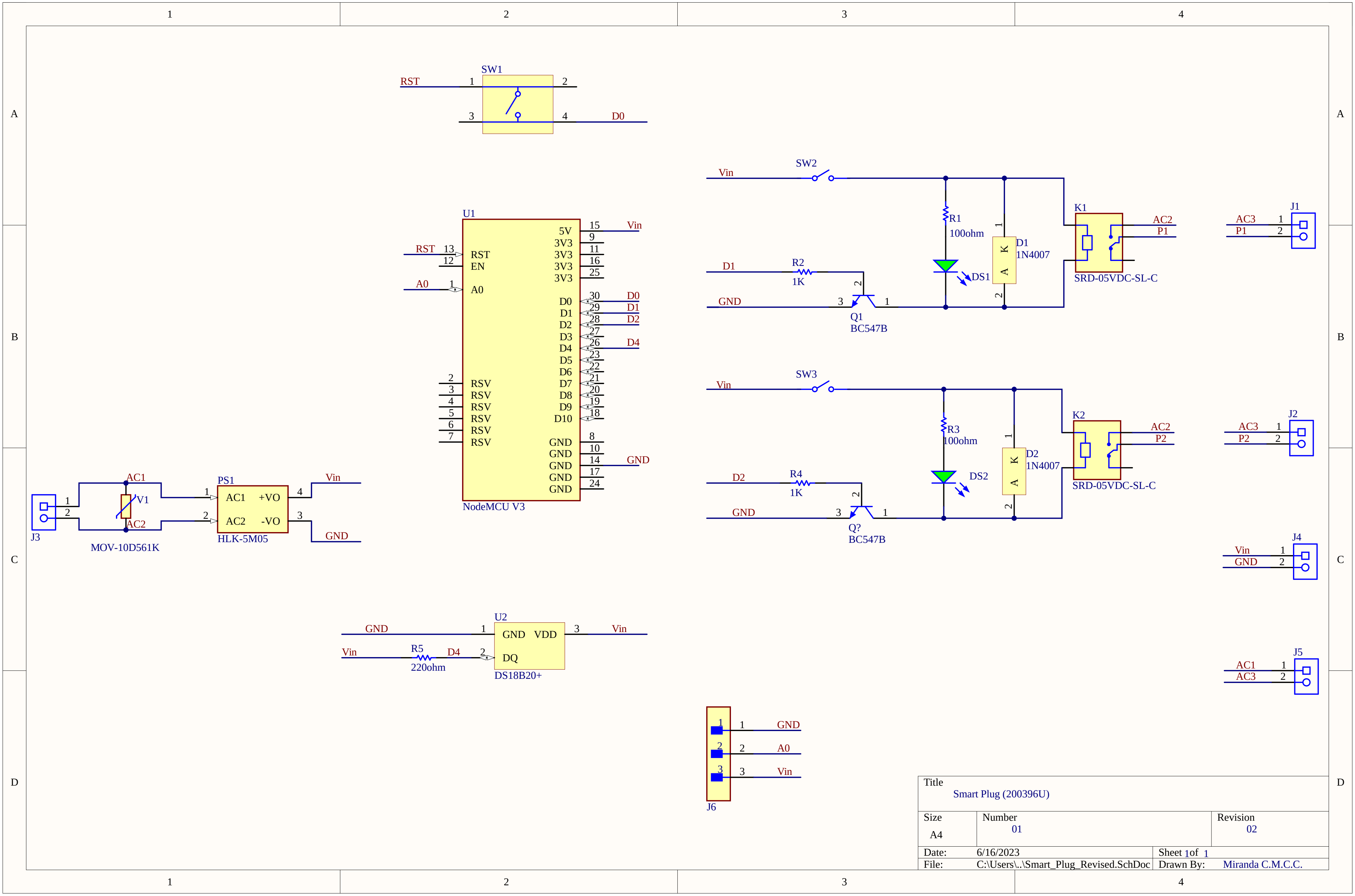


Figure 9: Block Diagram (Improved)

To accommodate the USB outlets in the schematic design, I incorporated the connector (J4). Additionally, to facilitate individual control of the two outlets, I introduced the rocker switches (SW2 and SW3) into the design.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RST
D0
U1
NodeMCU V3
5V
3V3
RST
EN
A0
D0
D1
D2
D3
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D5
D6
D7
D8
D9
D10
RSV
GND
Vin
GND
SW2
SW3
R1
100ohm
R2
1K
R3
100ohm
R4
1K
DS1
DS2
D1
D2
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Q1
Q?
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K1
K2
SRD-05VDC-SL-C
AC2
P1
P2
AC3
AC1
J1
J2
J3
J4
J5
J6
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-VO
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VDD
DQ
R5
220ohm
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Size
A4
Number
01
Revision
02
Date:
6/16/2023
Sheet 1 of 1
File:
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Drawn By:
Miranda C.M.C.C.